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Regional Anesthesia and Surgical Volume in Children Undergoing Nuss Repair: A Multicenter Review



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ABSTRACT

Introduction: Regional anesthetic approach and surgical volume have been shown to outcomes in patients undergoing Nuss procedure for pectus excavatum. However, their independent relationship is not described. We investigated how regional anesthesia and surgical volume are associated with length of stay (LOS), postoperative opioid use, operating room utilization, and complications.

Methods: This is a 9-center retrospective review of patients ≤ 21 ys after Nuss procedure for pectus excavatum (2016–2020). High-volume centers and surgeons defined as the upper-quartile for annual procedures. Outcomes were compared with mixed effects linear/logistic regression models with random intercepts by institution as appropriate. A single-center unadjusted analysis was performed of erector spinae catheter (ESC) utilization (due to nonuniform use; high-volume only at 1-center).

Results: 780 patients were included. Significant variation existed in anesthetic approach and volume. Cryoablation was independently associated with shorter LOS (–2.1 d; 95% confidence interval [CI]: –2.6, –1.7) and lower postoperative opioid utilization (–120 morphine milligram equivalents, 95% CI: –181, –58.1) but increased surgical time

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(+45 min; 95% CI: 30.3, 59.8). Individual surgeon volume was associated with decreased LOS (−0.3 d; 95% CI: −0.5, −0.01), though high-volume centers had increased complications (odds ratio 2.2; 95% CI: 1.1, 4.2). There was no association between anesthetic approach and surgical complications. Within the single center utilizing ESCs ($n = 138$), a shorter LOS (2 versus 3 d, $P < 0.01$) was observed compared to those not receiving an ESC ($n = 19$).

Conclusions: Analgesic approaches varied significantly across institutions and limited our ability to directly compare cryoablation and ESCs. On multivariate analysis, cryoablation was associated with decreased LOS and postoperative opioid use, irrespective of center and surgeon volume. ESCs were similarly associated with improved outcomes. Rigorous prospective comparison of ESCs and cryoablation is warranted.

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Introduction

Pectus excavatum is the most common chest wall deformity in children, occurring in approximately 1 in every 400 live births.¹ The minimally invasive Nuss procedure has become the favored operative approach among pediatric surgeons since it was originally described in 1998.² Substantial variability exists in the surgical and perioperative management of these patients. Perioperative pain management has evolved considerably over the years, most significantly and recently with the popularization of intercostal nerve cryoablation and erector spinae catheters (ESCs).^{3–5} These and other analgesic modifications have substantially decreased postoperative length of stay (LOS) and improved postoperative pain control in pectus patients.^{5–8}

Despite recent advancements and published improvements with various perioperative techniques (both surgical and anesthetic), there is still no universally accepted standardized approach, which likely contributes to variable outcomes across centers.² Anesthesia pain teams have multiple options for regional pain control, as well as different opioid-based strategies, including previously more commonly utilized Patient Controlled Analgesia (PCA) and epidural catheters.^{4,9–11} Outcomes and considerations such as LOS, postoperative opioid use, available expertise, time, and complications may guide which pain management approach to use.

Previous work has examined these outcomes based on center volume, specifically demonstrating improved outcomes at high volume centers^{12,13}; however, this study did not control for surgeon volume. Understanding the relative contributions of both center and surgeon volume, can shed light on the potential benefits of specialization. In addition, comparing outcomes of various pain management strategies for patients undergoing the Nuss procedure, would allow centers and surgeons to adjust their enhanced recovery protocols.

To address this knowledge gap, we performed a multi-institutional retrospective review to investigate the variability in anesthetic practices, impact of regional anesthetic approaches, and influence of both center and surgeon volume on postoperative outcomes. We hypothesized that substantial variability exists in the perioperative management of patients undergoing a Nuss procedure and that both increased surgical volume and regional anesthesia would be independently associated with improved outcomes compared to opioid-only or opioid-predominant strategies.

Materials and Methods

Cohort

After institutional review board approval (HUM00193647) with a waiver of consent at all participating sites, a 5-year retrospective chart review was conducted from 2016 through 2020 at 9 centers. Patients were included if they were less than or equal to 21 ys of age at operation and underwent a Nuss repair for pectus excavatum. Patients were identified using the International Classification of Diseases-9 (754.81, 754.82) and International Classification of Diseases-10 (Q67.6, Q67.7) codes and then manually reviewed by local study teams. Patients were excluded if they underwent any other surgical procedure or had pectus carinatum. Center and surgeon volumes were defined as “high-volume” if they were in the upper quartile of centers or surgeons performing Nuss procedures annually. This distinction was made to facilitate multivariate analysis controlling for volume at surgeon and center level. Eight centers completed the full 5-year review. One center had an overwhelmingly high volume, and due to the potential impact on data analysis, only 2 ys of data (January 2019–December 2020) were included from this center.

Outcome variables

The primary outcome for the study was LOS (days). Secondary outcomes of interest were major complications requiring intervention, surgical and operating room (OR) time (minutes), and postoperative oral opioid usage. Major 30-d complications were defined as death, bleeding requiring transfusion or procedural intervention, lung or cardiac injury, cardiopulmonary collapse requiring support, or any intervention such as a chest tube, reoperation, or unplanned bar removal or adjustment. OR time was defined as time from entering to leaving the OR, and surgical time was defined as incision start to incision close. Postoperative oral and IV push opioid usage was converted to morphine milligram equivalents (MMEs) for intermittent dosing. Data regarding PCA MME dosing was not included in this analysis secondary to limitations within the Electronic Medical Record and accurate data capture. Specifically, patient-controlled intravenous (IV) PCA versus Epidural PCA could not be delineated consistently as they were charted in aggregate within the pain management team record. Given our inability to discern the dosing levels, we elected to exclude this data from analysis. Of note, however, use of IV PCA was

documented and analyzed as binary data (yes/no) to better understand use in the total study population across centers. All centers performing cryoablation were completing bilateral ablation of five intercostal levels adjacent to bar placement during the study time period with the exception of a single center performing only 4 levels. ESCs employed an infusion of 0.2% Ropivacaine with patient dependent dosing.

Statistical considerations

Initial data processing included descriptive statistics of all 9 centers patient cohort demographics as well as center and surgeon volume, regional pain modalities utilized, and outcomes (LOS, OR time, surgery time, MME, and complications). Regarding regional pain modalities, percentages represent percent of total patients receiving a modality at a given center. As such, some centers demonstrate overlap of modalities, that is, cryoablation with additional local intercostal block. "Opioid-only" indicates the percentage of patients who only received opiate pain control (including oral, IV push, and/or IV PCA) in the absence of other modalities.

Traditional multivariable modeling would not adjust for significant interinstitutional variability in perioperative protocols, including patient counseling, nursing and ambulation policies, physical therapy involvement, operative techniques, and anesthesia structure. Preliminary screening for potential confounders was not performed; therefore, we used mixed effects modeling with random intercepts for each institution to adjust for variability between institutions. Multivariable mixed effects linear or logistic regression models were used as appropriate for continuous and binary outcomes, respectively.

Models included high surgeon and center volume, cryoablation, epidural catheter, local anesthetic intercostal block, and opioid-only pain management. For volume, centers and surgeons not considered high-volume served as the reference. For anesthetic approach, patients that did not receive that modality were combined as the reference; this was done to compare each approach against all other regional methods for clinical generalizability. ESCs were utilized at a single institution alone and not distributed across multiple institutions. As such this could not be included in the mixed models because the effect of that independent variable would be inseparable from the center-level effect. However, to address this, a separate single-center unadjusted univariate analysis of ESCs was performed to delineate potential associations with LOS and postoperative outcomes. Furthermore, an additional posthoc analysis compared the single high-volume single center ESC data with aggregate data from centers employing cryoablation. All analyses were completed using Stata (version 17; StataCorp, College Station, TX) with significance defined as a *P* value less than 0.05.

Results

Cohort and outcomes

During this 5-year study period, 780 patients met inclusion criteria. Of the total patient population 84% (*n* = 653) were

male, 93% (*n* = 726) identified as White, 6% (*n* = 48) as Hispanic, and 0.7% (*n* = 6) as Black. In addition, 1% (*n* = 8) identified as Asian American, 0.6% (*n* = 5) as Native American, and 4% (*n* = 32) race could not be identified. Median age at operation was 15.5 y (interquartile range [IQR] 14.3–16.8), and the median Haller index was 4.3 (IQR 3.6–5.4). Upper quartile for center volume was 23 Nuss procedures per year and this included 3 of 9 centers. The upper quartile for surgeon volume was 4 procedures per year and included 14 of 51 surgeons.

The choice of regional anesthetic was variable across and within most centers (Table 1). Cryoablation was used in four centers, epidurals in seven centers, local intercostal blocks in six centers, and ESCs in one center. Five centers had patients that were managed with opioids only without additional modalities. Furthermore, two centers had >70% of patients with opiates as the sole modality for pain control. Only two centers utilized one sole modality, while six centers utilized three or more modalities. Median OR time and median surgical time both demonstrated a wide range and were non-normally distributed (Table 1). Median LOS ranged from 1 to 5 d: specifically, 1 d (IQR 1–1) in those with cryoablation, 2 d (IQR 2–2) with ESCs, and 4 d (IQR 3–5) with epidurals and local intercostal nerve blocks. High-volume centers demonstrated shorter median LOS of 3 d (IQR 2–4) while low-volume centers had 4 d (IQR 3–4).

Median postoperative oral opioids ranged from 50 to 225 MME (Table 1), which did not include IV PCA dosing. Centers utilizing opioid-only analgesia demonstrated higher MMEs. Only 15% (*n* = 116) of the total study population of patients utilized an IV PCA during their hospitalization. Furthermore, 61 of these patients (53%) were treated with either opioids only or in conjunction with intercostal nerve blocks. The remaining patients with PCA's were in the epidural PCA group (*n* = 39, 33%) or cryoablation group (*n* = 16, 14%). Patients treated with ESCs at the single institution were not prescribed IV PCAs.

Major complications occurred in 81 (10%) patients (Table 2). The most common major category was an effusion or pneumothorax requiring intervention, occurring in 31 (4%) patients. Medical intervention (antibiotics) for complication was required in 31 (4%) and a surgical intervention in 60 (8%) patients. Patients who require surgical management of complications included tube thoracostomy in 30 (4%), reoperation for diagnoses other than infection in 26 (3%), and surgery for deep space infection in 18 (2%).

Surgical volume and regional anesthetics

Based on center and surgeon volume across sites, high surgeon volume was determined to be four or more cases per year and high center volume was determined to be 23 or more cases per year. The linear coefficients or odds ratios (where applicable) from multivariable modeling of center and surgical volume with regional anesthetic type are detailed in Table 3. Cryoablation was independently associated with a decreased LOS (−2.1 d; 95% confidence interval [CI]: −2.6, −1.7) and decreased postoperative opioids (−120 MME, 95% CI: −181, −58.1) but increased surgical time (+45 min; 95% CI: 30.3, 59.8). None of the other included anesthetic approaches were significantly associated with LOS, surgical time, or

Table 1 – Volume, regional pain modality, and secondary outcomes by site.

Site	Annual nuss volume	Cryo-ablation	Epidural	Local intercostal block	Erector spinae	Opioids only	Length of stay	Operating room time	Surgery time	Morphine milligram equivalents
1	78.5	-	8%	89%	88%	-	2 (2-2)	188 (172-212)	90 (77-108)	70 (53-125)
2	17.8	-	13%	5%	-	72%	4 (3-5)	167 (142-88)	88 (76-105)	225 (131-374)
3	4	-	100%	-	-	-	5 (4-5)	297 (257-332)	150 (115-181)	68 (41-161)
4	14	90%	9%	50%	-	-	2 (2-3)	194 (180-221)	130 (113-158)	72 (30-142)
5	23.2	55%	11%	62%	-	26%	2 (1-4)	179 (142-209)	122 (76-150)	64 (23-205)
6	7.6	42%	-	5%	-	32%	4 (3-4)	218 (193-245)	145 (116-167)	101 (60-203)
7	26.6	100%	-	-	-	-	1 (1-1)	135 (113-153)	95 (71-107)	50 (20-125)
8	10.2	-	10%	-	-	90%	4 (3-4)	135 (116-149)	75 (65-87)	202 (108-443)
9	21.2	-	84%	11%	-	5%	4 (3-5)	172 (155-194)	79 (64-96)	86 (60-166)

Values expressed as absolute value, percent of patients within the institution receiving a type of pain control, and median (interquartile range). Percentages reflect total patient numbers receiving modality at the given institution, as such overlap exists. "Opioids Only" indicates the percentage of patients receiving only opiate pain control in the absence of additional modalities (i.e., oral, IV Push and/or IV PCA). Length of Stay is in days, Operating Room and Surgery Times is in minutes.

Table 2 – Complications requiring intervention and the incidence of each subtype.

Major complications	81 (10%)
Effusion or pneumothorax requiring intervention	31 (4%)
Surgical site infection	16 (2%)
Mechanical bar problem requiring reoperation	15 (2%)
Bar infection	13 (2%)
Bleeding	5 (1%)
Pneumonia	5 (1%)
Pulmonary injury	2 (0.3%)
Cardiopulmonary failure	1 (0.1%)
Medical intervention: antibiotics	31 (4%)
Deep space infection requiring surgery*	18 (2%)
Superficial infection	15 (2%)
Pneumonia	3 (0.5%)
Surgical intervention	60 (8%)
Tube thoracostomy	30 (4%)
Reoperation	26 (3%)
Surgery alone for deep space infection	18 (2%)
Bar removal	5 (1%)

Includes 9 centers. Values expressed as absolute (percent) of total patients.

"Major Complications", "Medical Intervention," and "Surgical Intervention" count each patient (note, some had ≥1 subtype complication or intervention).

* Deep Space Infections (DSIs) Requiring Surgery and Antibiotic therapy.

postoperative opioid utilization, specifically epidural catheters, local intercostal block, or opioid-only dosing (Table 3).

High surgeon volume was associated with decreased LOS (-0.3 d; 95% CI: -0.5 , -0.01). Surgeon volume was not associated with surgical time, complications, or postoperative interventions. Higher volume centers had increased complications (odds ratio 2.2; 95% CI: 1.1, 4.2). There was no association between anesthetic approach and surgical complications or postoperative interventions. Post hoc univariate analysis comparing the single center utilizing ESCs and aggregate data from the 4 centers employing cryoablation demonstrated shorter LOS, lower MME, and shorter OR times in the cryoablation group (Table 4).

Discussion

In this multi-institutional review of the perioperative management of patients undergoing a Nuss procedure for pectus excavatum repair, surgical volume and anesthetic approach were associated with postoperative outcomes. Specifically, regional anesthesia with intercostal nerve cryoablation was independently associated with a shorter LOS and lower postoperative opioid utilization, and there was a small decrease in LOS among patients operated on by a high-volume surgeon. In addition, analysis of the high-volume single institution utilizing ESCs for regional pain management also demonstrated an association with a shorter LOS. Interestingly, an

Table 3 – Mixed effects modeling of surgical volume and regional approach on surgical outcomes.

Volume/ approach	Length of stay (d)	Operating room time (min)	Surgery time (min)	Morphine milligram equivalents	Major complications	Medical intervention	Surgical intervention
High center volume	-0.3 (-1.1, 0.4)	-68.7 (-138, 1.4)	-36.3 (-76.6, 4.1)	31.9 (-76.2, 140)	2.2 (1.1, 4.2)	1.9 (0.6, 5.8)	0.9 (0.4, 2.1)
High surgeon volume	-0.3 (-0.5, -0.01)	5.3 (-3.0, 13.5)	5.7 (-1.1, 12.5)	-4.4 (-38.5, 29.7)	0.5 (0.3, 1.1)	1.0 (0.3, 3.0)	0.8 (0.4, 1.7)
Cryoablation	-2.1 (-2.6, -1.7)	45.0 (30.3, 59.8)	47.9 (35.7, 60.0)	-120 (-181, -58.1)	1.2 (0.6, 2.4)	0.4 (0.1, 1.3)	0.6 (0.3, 1.3)
Epidural	0.2 (-0.3, 0.7)	3.5 (-13.6, 20.6)	3.0 (-11.1, 17.1)	54.5 (-15.8, 125)	1.1 (0.4, 3.1)	1.6 (0.3, 7.4)	0.8 (0.3, 1.4)
Local intercostal block	-0.4 (-0.8, 0.02)	-1.3 (-14.7, 12.0)	5.0 (-6.1, 15.9)	-20.1 (-74.5, 34.3)	1.1 (0.6, 2.3)	1.5 (0.4, 5.4)	1.5 (0.7, 3.3)
Opioids only	-0.3 (-0.7, 0.2)	-6.7 (-22.6, 9.2)	5 (-10.6, 15.7)	65.6 (-0.9, 132)	2.1 (0.8, 5.6)	0.8 (0.1, 4.2)	0.5 (0.2, 1.7)

n = 623, center #1 excluded due to primary use of erector spinae catheter. Results are from mixed effects linear or logistic regression models with random intercepts by institution. Values are reported as linear coefficient (95% confidence interval) for continuous outcomes and odds ratio (95% confidence interval) for binary outcomes. Significant values are bolded and defined as *P* < 0.05.

Table 4 – Comparison of surgical outcomes between ES catheters and cryoablation.

Outcome	ES Catheter (<i>n</i> = 138)	Cryoablation (<i>n</i> = 276)	<i>P</i> value
Length of stay (d)	2 (2-2)	1 (1-2)	< 0.01
Surgery time (mins)	90 (75-108)	118 (92-144)	< 0.01
Operating room time (mins)	191 (172-215)	173 (136-206)	< 0.01
Morphine milligram equivalents	65 (51-120)	48 (19-108)	< 0.01

Results displayed as median (interquartile range). Significant results bolded and defined as *P* < 0.05. ES = erector spinae.

increase in the frequency of major complications was seen at high-volume centers; however, this association was not present when examining high-volume surgeons.

Cryoablation has been shown to decrease postoperative pain and LOS^{3,6-8}; however, these prior studies did not adjust for center or surgeon volume. Our multicenter study demonstrates similar associations between cryoablation and decreased postoperative pain and LOS in these patients when also controlling for volume. There was no associated increase in postoperative complications or need for postoperative interventions between regional anesthetic approaches. Previous work in the adult population has demonstrated adverse effects and long-term outcomes of cryoablation including tissue breakdown, temporary and permanent peripheral nerve injury, and myonecrosis.¹⁴⁻¹⁸ Of note, this literature represents case reports, case series, and reviews largely in patients with chronic neuralgia, peri-operative management in those undergoing mastectomy, and traumatic rib fracture patients. Additional prospective studies in children and adolescents with long-term follow-up for potential complications associated with cryoablation are needed.

It is notable that the highest volume center in our cohort (center #1) did not use cryoablation but instead used ESCs with similarly low LOS (2 d) and postoperative opioid utilization (70 MME). Separating out the effect of ESCs from other systematic components of center #1's perioperative protocols is not possible, and therefore it is not clear whether utilization of ESCs would have comparable results at other institutions. Though a univariate analysis demonstrated a decrease in LOS, MME, and OR time for cryoablation compared to ESCs, it is important to note that this analysis has significant limitations in its comparison of single and multicenter data given an inability to control for confounders. Future rigorous studies comparing ESCs and cryoablation are warranted.

Pediatric surgery is a field encompassing many rare diseases, which makes defining the influence of surgical volume difficult.¹² In a large national database study looking at pectus excavatum repairs, high-volume centers were also defined using quartiles, and similarly volume greater than 26 cases per year for a center and greater than 10 for a surgeon were associated with the lowest risk of complications.¹³ However, this study included all procedures for pectus excavatum and did not differentiate between the Nuss and Ravitch

procedures. In our review, we did not include the Ravitch procedure given expected differences in postoperative pain and LOS, and instead focused solely on outcomes after a Nuss procedure.¹⁹⁻²¹ There are also significant technical differences between these two procedures and being high-volume in one does not necessarily translate to proficiency in the other. Contrary to what would be expected, high-volume centers were associated with a two-fold increase in major complications. While we do not have a clear explanation for this, it is possible that the highest volume centers have greater opportunity for complications given that they may receive referrals for and operate on cases of greater complexity, even compared to other tertiary pediatric centers. This unexpected finding warrants further investigation.

There are limitations to this study. As a retrospective review it is inherently susceptible to selection and underascertainment bias, although with 9 centers and 780 patients we provide data from a large sampling with improved generalizability compared to many prior single-institution studies. The final year of the study (2020) was during the COVID-19 pandemic, and hospitals had varying approaches to case triage based on local disease burden and resources.^{22,23} It is, therefore, possible an otherwise high-volume center could have fallen below the upper quartile if they limited Nuss procedures during that time. This study was not designed to look at changes in patient presentation or hospital triage systems, and hence we did not control for those complexities. In addition, we acknowledge that the inability to include IV PCA dosing underestimates the overall opioid dosing, though the number of patients this affected was less than 3% of the total study population. The number of bars placed during the procedure was not accounted for in operative time lending to potential bias in comparisons. Specific institutional management and recovery pathways were not documented for the purpose of this study. LOS was intentionally measured in days in order to control for potential effects of variances in care, such as social barriers to discharge or clinical/administrative challenges. Given the use of days, in lieu of minutes or hours, in addition to the large sample size of this data set, we do not anticipate these variances in care to have affected our results. Any examination of institution-level effects such as volume and regional anesthesia is also limited by the inability to control for all local confounding variables. However, we believe that the mixed effects modeling approach using the detailed clinical variables collected in this study reliably differentiates the specific contributions of surgical volume and regional anesthetic approach toward postsurgical outcomes in patients undergoing a Nuss procedure.

Conclusions

In this large multi-institutional review of analgesic approaches utilized for Nuss procedures in Pectus Excavatum significant variability in care was noted, including continued use of opioid-only methods. Our analysis supports and expands on previously published literature by demonstrating associated decreased LOS and decreased postoperative opioid use when employing cryoablation, irrespective of center and surgeon volume. ESCs were similarly associated with improved outcomes and represent another viable regional

analgesic approach. Centers and surgeons performing pectus excavatum repair via the Nuss procedure should consider standardization at the institutional level with respect to regional anesthetic approaches to decrease opioid usage in this population. Future prospective studies comparing ESCs to cryoablation as regional anesthetic strategy, as well as enhanced recovery pathways, could better delineate efficacy in postoperative pain control and outcomes.

Level of Evidence

IV.

CRediT authorship contribution statement

Dimitra Lotakis: Writing – review & editing, Writing – original draft, Validation, Resources, Project administration, Methodology, Investigation, Formal analysis. **Jack P. Vernamonti:** Writing – review & editing, Writing – original draft, Validation, Resources, Project administration, Formal analysis, Data curation. **Tiffany N. Wright:** Writing – review & editing, Project administration, Methodology, Investigation, Data curation. **Kyle J. Van Arendonk:** Writing – review & editing, Methodology, Formal analysis, Data curation. **Peter C. Minneci:** Writing – review & editing, Methodology, Investigation, Data curation. **Charles M. Leys:** Writing – review & editing, Methodology, Investigation, Data curation. **Matthew P. Landman:** Writing – review & editing, Methodology, Investigation, Data curation. **Shawn D. St. Peter:** Writing – review & editing, Methodology, Investigation, Data curation. **Rebecca L. Brown:** Writing – review & editing, Methodology, Investigation, Data curation. **Grace Z. Mak:** Writing – review & editing, Methodology, Investigation, Data curation. **Tsehay B. Abebe:** Writing – review & editing, Methodology, Investigation, Data curation. **K. Elizabeth Speck:** Writing – review & editing, Visualization, Validation, Supervision, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

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